

Blockchain – LDS7006M

Activity 1 (Session 5)- Understanding Blockchain Consensus Protocols (PoW and PoS)

Objective

To implement and compare the Proof of Work (PoW) and Proof of Stake (PoS) consensus protocols in a simplified blockchain simulation using Python.

Part 1: Proof of Work (PoW) Implementation

1. Block Structure

- Define a class `Block` that represents a block in the blockchain. Each block should contain attributes like `index`, `previous\_hash`, `timestamp`, `data`, `nonce`, and `hash`.
- Implement a method `calculate\_hash` to calculate the hash of a block.

2. Blockchain Implementation

- Create a class `Blockchain` that initializes with a genesis block.
- Implement a method `add\_block` to add new blocks to the blockchain using PoW.
- Implement a simple PoW mining function to find a valid nonce for each block.

Part 2: Proof of Stake (PoS) Implementation

3. PoS Block Structure (10%)

- Define a class `PoSBlock` that inherits from the `Block` class.
- Modify the block structure to include a `validator` attribute.

4. PoS Blockchain Implementation

- Create a class `PoSBlockchain` that initializes with a genesis PoS block.
- Implement a method `add\_block` to add new PoS blocks to the blockchain using PoS.
- Implement a PoS validator selection mechanism (e.g., based on stakes) to select validators for each block.

Part 3: Testing and Comparison

5. Testing

- Create instances of both the PoW and PoS blockchains.
- Add a few blocks to each blockchain.
- Ensure that the blocks are correctly added and mined using the respective consensus protocols.